LUNARSAFE // AUTONOMOUS HAZARD DETECTION METHODOLOGY

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STANDARD: ISRO Chandrayaan-3 LHDAC & NASA PDS4 Guidelines | Project Head: Darshan Patil

EXECUTIVE ABSTRACT

This technical document formalizes the autonomous lunar landing hazard detection, slope quantification, and candidate touchdown site selection methodology implemented in LunarSafe. Lunar landing safe zones require strict topography tolerances to avoid mechanical lander tip-over, chassis structural collapse, or propulsion plume re-impingement. LunarSafe establishes a fully deterministic, zero-hallucination processing pipeline capable of ingesting high-resolution planetary data from NASA LOLA digital elevation models (Float32 rasters) and ISRO Chandrayaan-2/3 Terrain Mapping Camera 2 (TMC-2) and Orbiter High Resolution Camera (OHRC) panchromatic products. The core engine applies Horn's (1981) third-order finite difference operator with 8*GSD stencil weighting for genuine physical slope quantification, accompanied by first-order analytical error propagation that models altimetric laser error (sigma_z = 1.0m) to enforce a conservative 2-sigma safety filter. Candidate terrain is partitioned into standard 24m x 24m ISRO landing grid cells, ensuring verifiable compliance with ISRO LHDAC mission thresholds (slope <= 10.0 deg, crater depth <= 1.0m, boulder height <= 0.30m). All computational decisions are bound to an immutable 5-stage SHA-256 evidence chain, guaranteed by Ed25519 digital signatures and AES-256-GCM encryption for complete post-flight mission auditability.

SECTION 1 — INTRODUCTION & CHANDRAYAAN-2 FAILURE ANALYSIS

The terminal descent of a lunar lander represents one of the most mission-critical phases in planetary exploration. In September 2019, the Chandrayaan-2 Vikram lander experienced an off-nominal trajectory deviation during its fine-braking phase. Subsequent failure analysis conducted by the National Expert Committee identified that tight thruster throttling limits combined with unexpected trajectory dispersion forced the guidance system to attempt rapid attitude corrections, resulting in high horizontal velocity at touchdown beyond the capability of the landing legs. This historic failure highlighted the paramount necessity of onboard autonomous real-time hazard detection with absolute determinism, wide safe-zone search radii, and mathematically bounded uncertainty margins.

LunarSafe addresses this challenge by providing pre-landing terrain qualification and onboard hazard map synthesis. Unlike deep learning black-box models that suffer from stochastic unpredictability and out-of-distribution hallucinations under varying solar illumination, LunarSafe relies on closed-form geometric algorithms that produce 100% bit-identical results across all computing platforms.

SECTION 2 — SYSTEM ARCHITECTURE OVERVIEW

The LunarSafe system is organized into a modular, decoupled architecture consisting of four integrated layers: (1) The Ingestion & Pre-Processing Engine, (2) The Frozen Deterministic Core (DeterministicLunarEngine), (3) The Uncertainty & Conservative Hazard Classifier, and (4) The Cryptographic Audit and Dossier Packaging Layer. This separation guarantees that core physical determinations remain completely isolated from telemetry visualization, network transport, or secondary reporting services.

SECTION 3 — THE 5-STEP DETERMINISTIC PROCESSING PIPELINE

Step 1: Input Ingestion	Streams raw IEEE 754 Float32 DEM elevation matrices or 16-bit PDS4 panchromatic rasters into contiguous buffers.
Step 2: Panchromatic Filter	Validates authentic grayscale lunar reflectance (< 0.1% chromatic pixels) to reject stock or corrupt data.
Step 3: Horn (1981) Slope Engine	Computes physical surface gradient using 3rd-order weighted 3x3 finite-difference stencils anchored to true GSD.
Step 4: Morphological Hazard Profiling	Identifies crater depression profiles and boulder highlight-shadow pairs via Sobel edge magnitude.
Step 5: ISRO 24m Grid & Audit Seal	Discretizes terrain into 24m cells, applies 2-sigma safety culling, and computes cryptographic hash chains.

SECTION 4 — INPUT DATA FORMATS (NASA LOLA & ISRO TMC-2)

1. NASA LOLA Gridded Data Record (GDR): Single-band IEEE 754 Float32 raster representing topocentric elevation in meters relative to the reference lunar sphere (R = 1737.4 km). Typical resolution ranges from 20.0m (polar tiles) to 59.225m (equatorial maria).

2. ISRO Chandrayaan-2/3 TMC-2 & OHRC: Planetary Data System version 4 (PDS4) standard format comprising an XML label (*.xml) and an uncompressed 16-bit Little-Endian (UnsignedLSB2) raw raster (*.img). Raster line and sample counts are parsed dynamically from the PDS4 label to prevent geometry mismatches across instruments. The bug was a dynamic-range assumption error. The loader divided raw pixel values by 65,535, but the sensor's actual dynamic range for this observation was only ~9–455. This crushed all pixels to near-zero, producing a blank image. The fix applies a 2-sigma dynamic-range stretch with gamma correction (gamma = 0.43), computed from the tile's actual mean and standard deviation.

SECTION 5 — PROCESSING PIPELINE SPECIFICATION

Contrast normalization is performed via physical exposure scaling without non-linear tone mapping. High-pass Laplacian unsharp masking enhances micro-relief boundaries and boulder ejecta rims without altering planar surface trends. The resulting gradient field is classified into safe, marginal, and hazardous terrain based on strict mission thresholds.

SECTION 6 — MISSION OUTPUT FORMATS

Vector PDF Mission Dossier	Multi-page flight briefing containing topography maps, safe zones, telemetry, and evidence hashes.
CSV Audit Matrix	Per-cell structured table reporting column, row, mean slope, sigma, safe/hazard flag, and coordinates.
JSON Telemetry Payload	Machine-readable stream with full module execution logs, timings, SHA-256 hashes, and A* rover trajectories.
Visual Hazard Maps	Calibrated 8-bit RGBA overlays indicating safe zones (green), slopes >10° (red), and culled marginal cells (yellow).